

# GIORGIO MORRA

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## INTRODUCTION

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Mechanical Engineering graduate student from Politecnico di Milano specializing in Mechatronics and Robotics, currently conducting Master's Thesis research at NASA JPL. I aim to contribute to innovative solutions for the control and navigation of Autonomous Systems.

## EDUCATION

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**M.Sc. in Mechanical Engineering (Mechatronics and Robotics)** 2023 – Ongoing

*Politecnico di Milano, Italy*

*International Mobility Extra-EU*: Pontificia Universidad Javeriana, Colombia

**B.Sc. in Mechanical Engineering** 2020 – 2023

*Politecnico di Milano, Italy*

*Erasmus+ Program*: Universidad Pontificia Comillas, Spain

## EXPERIENCE

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**Robotics Navigation Intern** Dec 2025 – Ongoing

**NASA Jet Propulsion Laboratory** – Robotic Mobility and Manipulation Section Pasadena, CA

Contributed to the **Venus Variable Altitude Aerobot** project, focusing on localization strategies for the Venusian atmosphere.

- Modeled the balloon's circumnavigation dynamic and interplanetary (Venus-Earth-Sun) communication;
- Simulated onboard measurements for sun sensor, altimeter, barometer and IMU;
- Designed the state prediction model and selected optimal filtering techniques;
- Performed sensitivity analyses to assess state estimation accuracy under varying sensor availability and noise levels;

## UNIVERSITY PROJECTS

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- UAV Autonomous Navigation in Urban Environment;
- Design, Simulation, and Dynamic Modeling of a 4-DOF SCARA Robot;
- License Plate Segmentation and Character Extraction;
- 1-DoF Robotic Exoskeleton for Elbow Rehabilitation.

## SKILLS

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**Prog. Languages:** Matlab, Python, C++, ROS2

**Simulation Tools:** Simulink, Simscape

**CAD Software:** Inventor, SolidWorks, Solid Edge, CATIA V5

**Various:** L<sup>A</sup>T<sub>E</sub>X, Microsoft Office

## LANGUAGES

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Italian (Native) — English (C1) — Spanish (C1)